



Cooperative navigation for UAVs in GNSS-denied area based on optimized belief propagation

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ABSTRACT

A multitude of individual unmanned aerial vehicle (UAV) cooperatively self-organized as a swarm attracts increasing attention for exploration and parallel tasks. Accurate position is essential for UAVs to avoid collision and implement control. Although the Global Navigation Satellite System (GNSS) has been integrated in UAV, it is susceptible to disruption. In this paper, we propose a novel cooperative navigation framework based on optimized belief propagation (BP) in the GNSS-denied area. Specifically, we derive a directional uncertainty model to characterize the position information of UAVs with different dynamic motions. With this model, we improve the navigation performance by designing a weighted uncertainty belief propagation (WUBP) based on information evaluation. Simulation and experimental results validate that in the dynamic scenarios, the proposed WUBP can make the swarm reap the benefits of more accurate positional estimate from UAVs with less maneuvers, higher-grade of INS or shorter time interval of GNSS signal lost.

1. Introduction

UAV swarms, inspired by swarms of animals in nature, are completely distributed and self-organized multiagent system [1]. These UAVs are ideal for completing tasks that can be broken into many parallel operations including disaster recovery, environment monitoring and military operations. Compared with the application of single UAV, a swarm system can offer multiple operation advantages in terms of cooperative effects, load capacity and cost effectiveness. Nevertheless, the swarm requires more emphasis on the accuracy of navigation information, which allows it to follow precise control commands and avoid collision while performing formation. In open airspace, stringent navigation capability is conventionally achieved with the aid of GNSS and inertial navigation system (INS). However, UAVs may lose GNSS signals while passing through the denied areas, which widely exist in operating environments such as urban canyons or in buildings. Fig. 1 shows a typical scenario of the GNSS-denied environments, in which the main problem is that the position error of INS tends to drift unbounded owing to the inertial sensor errors [2].

Alternatively, cooperation navigation is an emerging paradigm, in which UAVs help each other to improve navigation performance via fully exploit the relative relationship among all UAVs and the navigation information in a cooperative network. The main idea of cooperation navigation is illustrated in Fig. 1. When a team of UAVs (labeled 1 to 5) enter GNSS denied zone, although none of them can

independently maintain high-accuracy navigation, together they can cooperatively alleviate the growth of INS position error over time by fusing estimated signal metrics (such as Time of Arrival) and position information of neighboring UAVs. In addition, with the development of the Ultra-Wideband (UWB) in the field of wireless communication [3], high-precision relative measurements can be acquired at low cost. Thus, in this paper, we investigate the cooperation navigation method without external infrastructure to improve the navigation performance of UAV swarm.

In previous works, considerable efforts have been made to design appropriate algorithms for cooperative navigation. Classical algorithms such as extended Kalman filter (EKF) [4], particle filters (PF) [5], and optimization-based technique [6] are widely employed to solve the cooperative navigation problem. However, these algorithms are not amenable for a distributed implementation because they require a central processor to compute the agents' position, which makes it susceptible to single-node failure. Moreover, the communication bottleneck at the central processor and the curse of dimensionality problem become prohibitive to large team of UAVs. To address these weaknesses, recent literature [7–15] has focused increasingly on distributed solutions, which only require local information exchange to improve the location estimates. In [9], the proposed method utilizes simulated annealing to minimize the sum of square errors. The works in [10] and

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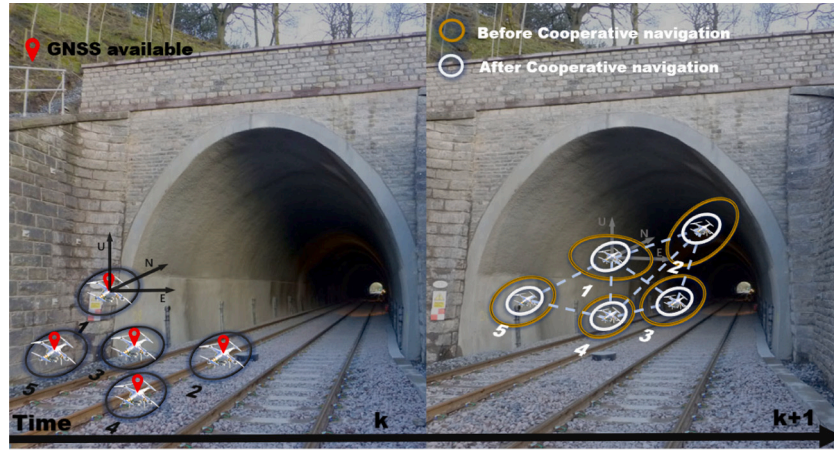


Fig. 1. Diagram of cooperation navigation for UAVs (the localization error is represented by an ellipse with the axis corresponding to the direction of localization error).

[11] are proposed to solve cooperative navigation problem by multi-dimensional scaling and map stitching. However, these methods only compute a deterministic estimate for agent. The absence of addition information about the reliability of estimates making these methods inappropriate for UAV swarm cooperative navigation, especially in the presence of INS errors.

To deal with above problem an algorithm known as belief propagation was introduced in [15], which is a popular member of Bayesian approach that runs across factor graphs. The effectiveness of BP-based algorithm is evidenced by its successful application in various fields [12–15] and will serve as the basic algorithm in this paper. However, most previous works involve surrounding anchors (such as well-equipped leader and fixed beacons) to provide precise knowledge about positions. In anchor-free environments, the accumulated error in positional estimate that caused by inertial measurement unit (IMU) severely hinders the development of Bayesian distributed algorithms in practical implementation. Solutions for cooperative localization in wireless sensor networks have been presented in [15] and [16], however, the system dynamic model is considered to be a displacement with additive Gaussian noise, which fails to hold in practice due to the neglect of accumulated errors caused by INS. The theoretical limit of localization accuracy was quantitatively analyzed in [1] and [16] by employing Cramer–Rao Bound, yet it mainly focuses on the positioning uncertainty induced by ranging link inaccuracy and is normally limited to off-line system analysis. Multiple-AUV cooperative localization without infrastructure was investigated in [13] and [14], with an understanding about the effects of inertial measurements, Yichen Li et al. designed an improved BP-based algorithm by considering a gradually increasing variance to approximate the accumulation of the position error. However, only limited improvement of performance is gained from this algorithm due to the neglect of dynamic factor. Besides, precise position information has a pivotal role in real-time flight control of UAVs for path tracking, whose uncertainty is often overlooked. Some effective control methods do include this information but in a disturbance tolerance way [17–19].

In spite of above research progress, a study particularly emphasizing on cooperative navigation based on dynamic uncertainty evaluation is still missing. In this paper, we exploit position uncertainty model by taking INS error statistics into account to improve the localization performance of UAV under dynamic scenarios. Specifically, the main contribution of this paper can be summarized as follows:

(1) We take real INS features into consideration and derive a position uncertainty model in terms of directions to characterize the position estimation, which shows relationship between the position uncertainty and the effective factors (e.g. quality of INS, dynamic maneuver, time interval of GNSS signal lost).

(2) A position information intensity is derived from inertial measurements statistics to reflect the quality of the estimate and adaptively tune the weights of navigation information based on the UAV position uncertainty model.

(3) A novel distributed cooperative navigation algorithm named WUBP based on Bayesian framework is proposed, in which the proposed directional position uncertainty model is employed to further reduce the impact of accumulated INS errors.

The rest of this paper is organized as follows. Section 2 describe the formulation of the problem to be solved and system framework. The expression for the directional position uncertainty model is derived in Section 3. Section 4 presents the proposed algorithm. Sections 5 and 6 validate the proposed method through numerical simulation and real world experiment. Finally, a conclusion is drawn in Section 7.

2. Problem formulation and system framework

2.1. Problem formulation

Consider a team of N_a (labeled 1 to N_a) UAVs navigating in GNSS-denied areas without infrastructure, each UAV is equipped with on-board IMU sensor and UWB radio devices for intra- and inter-nodes measurements. A typical scenario is shown in Fig. 1, in which the members of UAV team exchange navigation information with neighbors so as to improve their individual position estimates. Both the measurement and calculation processes are made in discrete time instants t_n ($n = (1, 2, \dots, N)$). The position of UAV i at time t_n is described by $x_i^{(t_n)}$. Within the team, the neighbor set of UAVs from which UAV i can receive and decode transmission at time t_n is denoted by $\Gamma_i^{(t_n)}$. Using packets from $j \in \Gamma_i$, UAV i is able to make measurements of signal metrics (denoted by $z_{j \rightarrow i}$), which contain information about relative distance and bearing between UAVs. We denote the set of relative measurements by $z_{rel}^{(1:t_N)} = \{z_{j \rightarrow i}^{(1:t_N)}\}_{j \in \Gamma_i, i=1, \dots, N_a}$. Additionally, UAVs utilize local inertial measurements (denoted by z_{self}) to estimate its own position between different timesteps. The full set of measurements is denoted by $z^{(1:t_N)} = \{z_{self}^{(1:t_N)}, z_{rel}^{(1:t_N)}\}$.

The objective of cooperative navigation in GNSS-denied areas is to determine UAV positions from prior knowledge as well as measurements with errors, and further alleviate the impact of accumulated errors induced by IMU. From a Bayesian perspective, this task means to determine joint posteriori probability distribution function (PDF) $p(x_{1:N_a}^{(0:t_N)} | z^{(1:t_N)})$, given measurements up to time t_N . Note that only horizontal position of UAV is considered due to the barometer can be used to measure altitude.

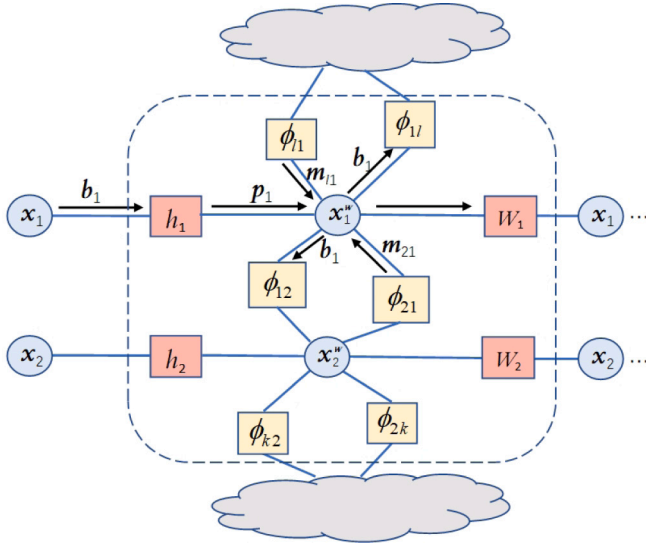


Fig. 2. Cooperative network factor graph of the proposed cooperative navigation framework. We omit the superscript of time slot for simplicity and the clouds represent the remainder of the network factor graph. $p(z_{j \rightarrow i}|x_i, x_j)$ and $p(x_i|x_j, z_{self})$ are abbreviated by ϕ_{ji} and h_i , the proposed weighting function is denoted by W_i .

2.2. System framework

In this paper, we develop a Bayesian framework for UAV swarm cooperative navigation while considering the position estimation uncertainty and information reliability, which incorporates a factor graph with the proposed weighting function and particle-based BP algorithm.

Cooperative navigation problem can be mathematically formulated as an inference problem. According to Bayes' Rule and conditional independence, the joint posteriori PDF $p(x_{1:N_a}^{(0:t_N)}|z^{(1:t_N)})$ can be decomposed into:

$$\begin{aligned} & p(x_{1:N_a}^{(0:t_N)}|z^{(1:t_N)}) \\ & \propto p(x_{1:N_a}^{(0)}) \times \prod_{i=1}^{t_N} \{p(x_{1:N_a}^{(i)}|x_{1:N_a}^{(i-1)}, z_{self}^{(i)}) \times p(z_{rel}^{(i)}|x_{1:N_a}^{(i)})\} \\ & \propto \prod_{i=1}^{N_a} p(x_i^{(0)}) \times \prod_{i=1}^{t_N} \prod_{j=1}^{N_a} \{p(x_i^{(i)}|x_i^{(i-1)}, z_{self}^{(i)}) \times \prod_{j \in I_i^{(u)}} p(z_{j \rightarrow i}^{(i)}|x_i^{(i)}, x_j^{(i)})\} \end{aligned} \quad (1)$$

where $p(x_i^{(0)})$ is the a prior on the UAV's initial position, and $p(x_i^{(i)}|x_i^{(i-1)}, z_{self}^{(i)})$ is the mobility model, $p(z_{j \rightarrow i}^{(i)}|x_i^{(i)}, x_j^{(i)})$ denote the model of UWB relative measurement.

However, to obtain the minimum mean-square-error (MMSE) estimate of $x_i^{t_N}$, direct marginalization of the posterior PDF $p(x_{1:N_a}^{(0:t_N)}|z^{(1:t_N)})$ is unfeasible due to the total computational amount grows with the dimension of state space, which known as the curse of dimensionality problem [20]. BP is an iterative message-passing algorithm based on the theory of factor graphs [21] whose calculation complexity scales only linearly with the time steps. Therefore, it is attractive for cooperative navigation problem.

We map the factor graph onto the physical network topology, resulting in a cooperative network factor graph as shown in Fig. 2. The main modification in the proposed framework is shown in the dotted box. The circle node x and x^w denote variable with respect to position estimate and estimate with uncertainty. The square node indicates a function of the variables connected to it. In cooperative network factor graph, we refer to the marginal posterior distribution of UAV position and the information transmitted from one UAV to another as belief and message respectively. The arrows above the edges indicate the information passing directions. The basic operations performed on the factor graph can be found in [15].

3. Derivation of directional position uncertainty model

During UAV navigate in GNSS-denied areas, the position estimates associated with predicted belief tends to diverge with time due to unhindered drift in inertial sensor measurements. Having more than 160 m position error after 60 s can be typical (with tactical grade sensor). Thus, using the beliefs produced by prediction step directly often causes BP algorithms to behave unpredictably in terms of accuracy and even lead it to fail. In this section, we address these concerns by analyzing the statistic properties of INS accumulated position errors, leading to the closed-form expressions of the position estimation uncertainty model, which can characterize the predicted belief and act as a navigation performance indicator.

3.1. Error model of inertial system

UAV onboard INS is a nonlinear and time-varying system, whose error is unbounded and closely related to the specific navigation trajectory of the carrier. Thus, it is hard to obtain an accurate analytical solution of the INS dynamic error model. To characterize the position errors induced by INS and mitigate the defect of BP-based algorithm failure, we simplify the complete INS error model out of consideration for UAV carrier maneuvers, leading to the time-domain solution of simplified system position error differential equation. Details of solution procedure are provided in Appendix.

$$\delta p(t) = \phi(t)X(0) + \int_0^t \phi(t-\tau)Bd\tau \quad (2)$$

where δp is the error vector of position, $X(0)$ is the initial error state, B denotes the sensor error, $\phi(t)$ is the state transition matrix.

It is evident that the position errors drift with time due to the errors of accelerometers and gyroscopes integrated in the INS. In addition, since the initial error state can be accurately estimated when UAV aided by GNSS, its influence can be safely neglected. Hence, it is necessary to link the statistical characteristic of sensor error model to that of position estimation uncertainty.

The inertial sensor errors are mainly gyroscope drifts and accelerometer biases that incorporate deterministic as well as non-deterministic components. The deterministic components include residual scale factor errors, which characterize non-random effects and introduce errors proportional to the sensor measurement. Moreover, the stochastic components of the sensor errors are random and usually contain several error sources. These errors are usually correlated with time, in this paper, only static bias, dynamic bias and angle random walk noise are taken into account due to they have the strongest influence on INS performance. For gyroscope and accelerometer, the measurement error model can be expressed as

$$\delta \theta = \delta K\theta + e_b + e_r + w \quad (3)$$

where $\theta = [\omega_{ib}^b \ f_{st}^b]^T$ denote the measured angular rate and acceleration, $\delta \theta$ is the corresponding measurement error, $\delta K = \text{diag}\{\delta K_G \ \delta K_A\}$ is the deterministic part of error and $e_b = [\epsilon_b \ \nabla_b]^T$ denote the static bias. w is the white noise with variances σ_w^2 , $e_r = [\epsilon_r \ \nabla_r]^T$ represent the dynamic bias defined as

$$\dot{e}_r = -\tau e_r + w_r \quad (4)$$

where $\tau = \text{diag}\{1/\tau_g \ 1/\tau_a\}$ denote the correlation time associated with the Gauss-Markov processes, w_r is the white noise with variances σ_r^2 .

By considering the dynamic bias as a stationary random process, the mean and variance of sensor error model can be written as

$$E\{z(t)\} = \delta K\theta + (I + \exp(-\tau t))e_b = \delta K\theta + e \quad (5a)$$

$$D\{z(t)\} = \sigma_r^2 \tau / 2 + \tau_w^2 = \sigma_{\delta \theta}^2 \quad (5b)$$

where $e = [\epsilon \quad \nabla]^T$ is the sensor bias, and $\sigma_{(\delta\theta)}^2 = [\sigma_{(\delta\omega)}^2 \quad \sigma_{(\delta f)}^2]^T$ denote the variance associate with stochastic error. The quality of inertial measurements is determined by these parameters, which were set different values to characterize different grade of INS in the simulation.

In addition, owing to the flight endurance for UAV is short and directly related to the capacity of the battery, we simply suppose that the time of UAVs entering GNSS-denied area is less than 1/4 Schuler oscillation period of 84.4 min, thus the trigonometric function can be simplified using small angle approximation. Note that the position error equations with respect to the north and the east direction are in similar form, we only consider the north position channel for simplicity. By taking into account Eqs. (A.4a) and (A.4b), the statistical characteristics of position estimation uncertainty then can be obtained as follows

$$E\{\delta p_N(t)\} \approx -\frac{t^3 g}{6}(\delta K_{Gx} \cdot \omega_x + \epsilon_x)C_b^n - \frac{t^3 f_E}{6}(\delta K_{Gz} \cdot \omega_z + \epsilon_z)C_b^n + \frac{t^2}{2}(\delta K_{Ay} \cdot f_y + \nabla_y)C_b^n \quad (6a)$$

$$D\{\delta p_N(t)\} \approx \frac{t}{2}(3\sigma_{(\delta\omega_x)}^2 R^2 - \frac{3\sigma_{(\delta\omega_z)}^2 R^2 f_E^2}{g^2} + \frac{\sigma_{(\delta f_y)}^2}{w_s^2})C_b^n - t(2\sigma_{(\delta\omega_E)}^2 R^2 - \frac{2\sigma_{(\delta\omega_z)}^2 R^2 f_E^2}{g^2})C_b^n - t(\frac{\sigma_{(\delta f_y)}^2}{2w_s^2} - \frac{\sigma_{(\delta\omega_x)}^2 R^2}{2} + \frac{\sigma_{(\delta\omega_z)}^2 R^2 f_E^2}{2g^2})C_b^n \quad (6b)$$

where the subscript x , y and z denote the coordinate components in body frame.

It is obvious that both the mean and variance of position uncertainty increase with time. In particular, the mean value is mainly affected by the deterministic parts of sensor measurements, which result in uncertainty proportional to time cubed and squared respectively. Same interest is the stochastic parts of sensor errors as they introduce the variance of position uncertainty grows with $\sqrt{t}$.

3.2. The position uncertainty model

To properly model the position errors, we consider the sum of the mean and standard deviation of the position estimation uncertainty as a whole, and propose a directional position uncertainty model as describes below

$$U_{pN}^{(t)} = E\{\delta p_N(t)\} + \sqrt{D\{\delta p_N(t)\}}. \quad (7)$$

The Eq. (8) gives the INS-based position estimation uncertainty as a function of three different type of factors. The first type is dynamic characteristics of UAV, which would stimulate the residual scale factor errors and introduce a position uncertainty proportional to the rotation and acceleration rate. The second factor is the precision of inertial sensor. Usually the inertial sensors tend to dictate the quality of IMU and thus the accuracy of algorithm output. Finally, the third factor is the time interval of GNSS signal lost. Given the same quality INS, the UAV enters the GNSS-denied area for a much shorter duration has significantly smaller position uncertainty than other UAVs.

Furthermore, an agent in cooperative navigation network can be thought of as an anchor when the directional position uncertainty is zero along any direction (e.g., when UAV aided by map matching technology). To provide the insights into the connection between theoretical and algorithmic perspective, given the same grade of INS and duration of maneuver, a visual illustration of dynamic position uncertainty is shown in Fig. 3.

As can be seen in Fig. 3, the position error not only depends on the performance of the sensors but also includes dynamic effects. The true position error value is fluctuating according to the acceleration and turning rate. The model based approximated error can adapt the real position error changes with the same tendency. Note that the proposed directional position uncertainty model can only approximate the accumulated errors under short-term condition, due to the neglect of coupling term and small angle approximation.

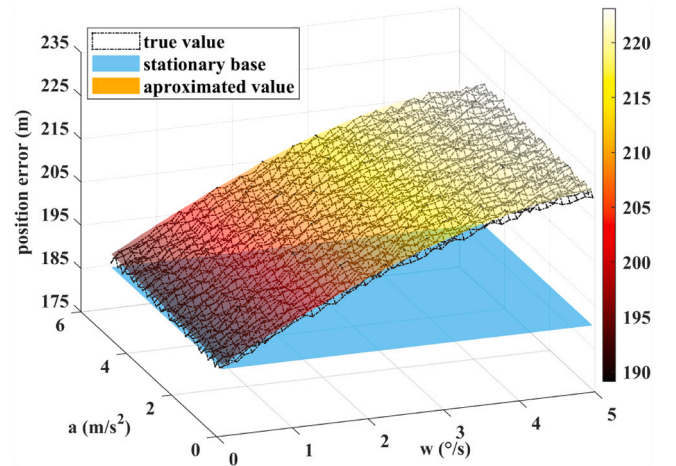


Fig. 3. The comparison of the true error value with 3000 samples (black dash net) and the approximated value (colored surface) from directional position uncertainty model. The blue surface indicate error on a stationary base.

4. UAV swarm cooperative navigation algorithm

The directional position uncertainty model is not only helpful to assess the localization accuracy, it can also be fed into a decentralized Bayesian cooperative navigation algorithm to improve the prediction accuracy online. In this section, we propose a weighted uncertainty belief propagation algorithm termed as WUBP to alleviate the impact of IMU errors, which is based on the theory of factor graphs and the sum-product algorithm (SPA). The proposed algorithm aims to slow down the position error growth and improve the localization accuracy under GNSS-denied environment.

According to the update rule of message passing [21] on a factor graph, the whole process of proposed WUBP algorithm for each UAV can be summarized by three key steps. In the following, we will introduce them in detail.

4.1. Design of the position prediction

For the first step, a UAV predicted belief is computed based on its positional estimate at the last timestep and mobility model $p(x_i^{(t)} | x_i^{(t-1)}, z_{self})$ combined with position uncertainty. Suppose that the predicted belief of UAV at time t is represented by a finite set of particles with associated weights $b_{pre}(x_i^{(t)}) = \{x_{pre,k,i}^{(t)}, w_{pre,k,i}^{(t)}\}_{k=1}^R$. As barometers are used to ensure the UAVs maintain a stable altitude, we assume the UAVs navigating in planes of different altitude. Each UAV can only predict their position $x_{pre,i}^{(t)}$ with INS

$$x_{pre,i}^{(t)} = \hat{x}_i^{(t-1)} + f(\Omega) \quad (8)$$

where $\hat{x}_i^{(t-1)}$ is the position estimate at the last timestep, and $f(\Omega)$ is the position increment of INS.

Then a position uncertainty matrix $U_i^{(t)} = \text{diag}\{U_{pN,i}^{(t)}, U_{pE,i}^{(t)}\}$ is calculated according to Eq. (8). Consequently, the predicted belief $b_{pre}(x_i^{(t)})$ can be obtained by directly drawing R independent particles from distribution $N(x_{pre,i}^{(t)}, U_i^{(t)})$ with equal weight. The position uncertainty matrix $U_i^{(t)}$ is employed to approximate the inertial navigation accumulated errors with respect to main influence factors. Such a design is able to compensate the inertial measurement errors. Fig. 4 exhibits the basic idea of the proposed position prediction step.

In Fig. 4, the red dots in Zone 1 denote the past belief distribution. The predicted position given by INS is represented by the magenta particles in Zone 2, which obviously do not include the real position. The state-of-the-art method in [14] address this problem by designing

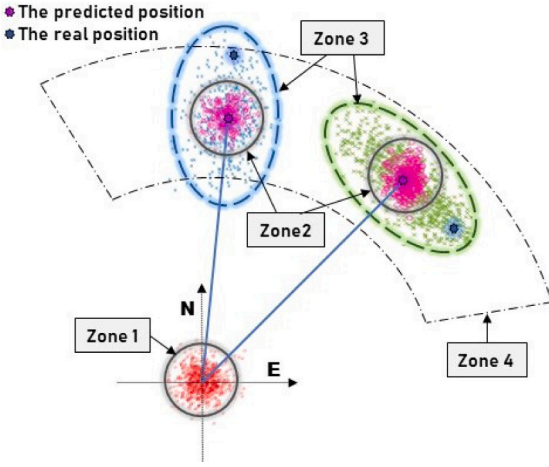


Fig. 4. The position prediction method is able to filter particles near the real position in Zone 3.

a gradually increasing variance to approximate the accumulation of the measurement errors, which is able to make the particles of the prediction (Zone 4) always cover the true position. However, it is failed to determine the position estimation uncertainty for different influence factors and to correctly represent the probability density distribution, leading to limited performance improvement.

Alternatively, given an initial position distribution and sensor measurements, the proposed prediction method is able to characterize the position error evolution and covers the true position with a much smaller area termed as Zone 3, in which the blue and green ellipse denote position estimation uncertainty caused by different maneuvers. The key point of our design is to cover the real position area with higher particle density by simulating the error accumulation of UAV position, it is vital to obtain desirable positional estimates. In addition, the proposed method is more robust when the UAV maneuvers, which can be ascribed to the consideration of dynamic error characteristics. With the information fusion operation introduced later in the third step, the particles near the true position in Zone 3 can be filtered out.

4.2. Message computing

The next step is to compute the incoming messages. According to the correction step of BP, the incoming messages can be interpreted as relative observation of UAV from its neighbor, which are related to the relative signal metrics and predicted beliefs of neighbors. Note that only a set of parameters in the Gaussian distribution are transmitted to reduce the communication cost. In the proposed algorithm, we set the number of messages passing iterations to one due to each UAV is able to obtain a prior knowledge of position based on its own inertial measurements. Consequently, the incoming messages can be rewritten as:

$$m_{j \rightarrow i}^{(t)}(x_i^{(t)}) \propto \int p(z_{j \rightarrow i}^{(t)} | x_i^{(t)}, x_j^{(t)}) \cdot b_{pre}(x_j^{(t)}) dx_j^{(t)}. \quad (9)$$

The incoming messages can be interpreted as relative observation of UAV from its neighbor, which is represented by a set of parameters from distribution $N(x_j^{(t)}, U_j^{(t)})$ to reduce the communication cost. To obtain a particle representation $\{x_{m,k,j \rightarrow i}^{(t)}, w_{m,k,i}^{(t)}\}_{k=1}^R$ of the incoming message $m_{j \rightarrow i}^{(t)}(x_i^{(t)})$, we first draw particles $\{x_{b,k,i}^{(t)}, w_{b,k,i}^{(t)}\}_{k=1}^R$ from distribution $N(x_{pre,j}^{(t)}, U_j^{(t)})$ as the representation of the received belief. As for the distance measurements obtained by the radio signal metrics $p(z_{j \rightarrow i}^{(t)} | x_i^{(t)}, x_j^{(t)})$, we draw particles of measured inter-node distance $\hat{d}_{j \rightarrow i}^{(t)}$ from a Gaussian distribution $N(\hat{d}_{j \rightarrow i}^{(t)}, \sigma_{d,j \rightarrow i}^2)$, as well as particles of angle

$\phi_k^{(t)}$ from a uniform distribution on $(0, 2\pi]$. Note that the weights of particles remain unchanged due to the cooperative estimate has not been computed yet. Each particle of incoming messages is then given by:

$$x_{m,k,j \rightarrow i}^{(t)} = x_{b,k,i}^{(t)} + d_k^{(t)} [\sin(\phi_k^{(t)}); \cos(\phi_k^{(t)})]. \quad (10)$$

4.3. Design of the PII and location estimate

The third step is to update belief by fusing information from the position prediction with incoming messages. Since each incoming message contains positional estimate with different uncertainty, treating these messages in equal can easily lead to inaccurate estimates. To suppress this adverse impact, a nonnegative number called the position information intensity (PII) is proposed, which is able to characterizes the uncertainty of position and tune the weights of incoming messages for information fusion.

In the UAV cooperative navigation system, upon the arrival of the broadcasted packets from neighbors $j \in \Gamma_i$, the position information intensity corresponding to UAV can be expressed as:

$$w_{ij}(t) = \begin{cases} 1 + (J_i\{p(t)\} - J_j\{p(t)\})/c & \text{if } J_i\{p(t)\} > J_j\{p(t)\} \\ J_i\{p(t)\}/J_j\{p(t)\} & \text{if } J_i\{p(t)\} \leq J_j\{p(t)\} \end{cases} \quad (11)$$

where c is the scaling factor with respect to IMU quality, $J_i\{p(t)\}$ denote the integrated position uncertainty of UAV i , which can be derived as:

$$J_i\{p(t)\} = \sqrt{(U_{PN,i}^{(t)})^2 + (U_{PE,i}^{(t)})^2}. \quad (12)$$

After all position information intensity of UAV i have been calculated, they are normalized by their sum. The introduced position precision coefficient $w_{ij}(t)$ takes advantage of different impact factors to balance the contribution of incoming messages. More specifically, incoming messages with greater uncertainty are given a lower value, and vice versa. Given the particle representation of position prediction and incoming messages, kernel density estimation (KDE) [22] is applied here to translate the particles into distribution. For a particle set $\{x_k, w_k\}_{k=1}^K$ of a distribution $p(x)$, and the KDE of $p(x)$ can be obtained as:

$$\hat{p}(x) = \sum_{k=1}^K w_k K_\sigma(x - x_k) \quad (13)$$

where $K_\sigma = \frac{1}{2\pi\sigma^2} \exp(-\frac{\|x\|^2}{2\sigma^2})$ is the Gaussian kernel with bandwidth σ . The bandwidth σ is determined by the ranging measurements [23], which is crucial for the performance of the estimate.

Let $\hat{b}_{pre}(x_i^{(t)})$ and $\hat{m}_{j \rightarrow i}^{(t)}(x_i^{(t)})$ be the KDEs of predicted belief and the incoming messages, respectively. Then the belief can be updated by:

$$b(x_i^{(t)}) \propto \hat{b}_{pre}(x_i^{(t)}) \prod_{j \in \Gamma_i^{(t)}} w_{ij}(t) \hat{m}_{j \rightarrow i}^{(t)}(x_i^{(t)}). \quad (14)$$

Due to the difficulty of directly sampling from target distribution $b(x_i^{(t)})$, importance sampling method [24] is used here to address these difficulties. With the principles of the importance sampling, we draw equal-weight particles from a distribution $q(x)$ that is easy to sample from, and then reweight all the particles by:

$$v_{b,k,i}^{(t)} \propto \frac{\hat{b}_{pre}(x_{b,k,i}^{(t)}) \prod_{j \in \Gamma_i^{(t)}} w_{ij}(t) \hat{m}_{j \rightarrow i}^{(t)}(x_{b,k,i}^{(t)})}{q(x_{b,k,i}^{(t)})} \quad (15)$$

where distribution $q(x) = \sum_{j \in \Gamma_i^{(t)}} \hat{m}_{j \rightarrow i}^{(t)}(x_{b,k,i}^{(t)})$ is a mixture of the incoming messages. After normalization and resampling operations, the belief of UAV i could be represented by the particle set $B_i^{(t)} = \{x_{b,k,i}^{(t)}, v_{b,k,i}^{(t)}\}_{k=1}^K$, which is an approximation of the posteriori distribution. Consequently, the particles near the real position are filtered out from the prediction with the incoming messages.

Table 1

Simulation settings.

Parameter	Value	Parameter	Value
Simulation time	100 s	Initial velocity	2 m/s
Gyroscope drift ϵ	1°/h	Accelerometer bias ∇	1 mg
Noise of gyroscope $\sigma_{(dw)}$	0.1°/h	Noise of accelerometer $\sigma_{(df)}$	0.1 mg
Correlation time τ	300 s	Scale factor error	100 ppm
Ranging standard deviation	0.2 m	Particle number M	500

Hence the minimum mean square error (MMSE) estimate of UAV can be provided by the mean of the posterior distribution as follows:

$$\hat{x}_{MMSE,i}^{(t)} = \sum_{k=1}^K v_{b,k,i}^{(t)} x_{b,k,i}^{(t)}. \quad (16)$$

Algorithm 1 Outline of WUBP Algorithm at Timestep t

- 1: **for** each UAV $i \in [1, N]$ **do**
 - 2: compute position estimate $x_{pre,i}^{(t)}$ and uncertainty matrix $U_i^{(t)}$ according to Eq. (8) and Eq. (7)
 - 3: broadcast predicted belief $b_{pre}(x_i^{(t)})$
 - 4: receive beliefs from neighbors $j \in \Gamma_i$ and extracts signal metrics $z_{j \rightarrow i}^{(t)}$
 - 5: compute incoming messages $m_{j \rightarrow i}^{(t)}(x_i^{(t)})$ according to Eq. (10)
 - 6: compute and normalize the position information intensity according to Eq. (11)
 - 7: draw equal-weight particles from distribution $q(x)$ and compute corresponding weights according to Eq. (15)
 - 8: get $B_i^{(t)}$ by normalizing the weights and resampling
 - 9: compute the MMSE position estimate $\hat{x}_{MMSE,i}^{(t)}$ according to Eq. (16)
 - 10: **end for**
-

5. Simulations

In this section, we evaluate the performance benefits of the proposed methods by comparing with the state-of-the-art methods in two simulation scenarios: the dynamic maneuver scenario and the cruise scenario. The inputs to the simulation are simulated measurements and noises obtained from the inertial sensors. In the simulation, the impact of several key parameters was examined.

5.1. Simulation setup

Consider a team of UAVs enters GNSS-denied zone with an initial position error of 0.5 m and thereafter entirely counts on cooperative navigation to mitigate the INS error growth. UAVs are equipped with both IMU and UWB, and their sampling rates are set to be 10 Hz and 1 Hz respectively. To keep consistency with the setup in the real experiment, noise characteristics of IMU are based on parameters taken from real sensors. The ranging and bearing observations are provided by UWB module with errors following zero-mean Gaussian distribution. We setup scenarios with different generated UAV trajectories in Fig. 5, and the root mean square error (RMSE) is considered as the performance metric.

5.2. Effect of maneuvers on RMSE

To demonstrate the performance gain provided by the exploitation of the directional position uncertainty model and position information intensity, WUBP algorithm and the state-of-the-art methods are compared. In Fig. 5(a), the UAV team traveling directly North under three operating conditions: UAV 1, 2 and 5 follow a smooth flight path; UAV 3 performs dynamic maneuvers including acceleration and deceleration; UAV 4 performs spiral maneuver. The parameters of simulation conditions are depicted in Table 1. We evaluate the algorithms using

Monte-Carlo simulation with 100 times. The RMSE against the time of navigation is shown in Fig. 6. We also validate the performance of these methods on different dynamic conditions as shown in Fig. 7. The acceleration and rotation velocity of UAV 3 and 4 in Fig. 6(a) is chosen to be (0.5, 1), (1, 2), (2, 4), (4, 6). The following key observations can be obtained:

(1) All of three algorithms can slow down the INS error divergence, while the proposed algorithm gives the best localization accuracy. The inferior performance of SPBP [23] algorithm mainly results from the neglect of accumulated inertial measurement errors in the system dynamic model. Consequently, the particles from predicted beliefs are placed outside the region of true position and the resulting performance is poor.

(2) In the BPUCL [14] algorithm, an accumulated variance is designed to simulate the error accumulation of the inertial measurement, which improve the predicted beliefs notably. However, the BPUCL algorithm is unable to properly characterize UAV position estimation uncertainty and qualify the reliability of information provided by different UAVs, especially when the position errors grow large.

(3) It can be observed from Fig. 7 that the dynamic maneuver significantly produces negative effect on the localization accuracy. The best accuracy is obtained with the proposed WUBP method, meanwhile it outperforms the others BP-based methods in terms of robustness. The reason for this can be explained as follow: since the proposed directional position uncertainty model can approximate the short-term theoretical accuracy with the effective factors, the impact of dynamic maneuver is alleviated in the proposed algorithm. In addition, owing to the utilization of the position information intensity based on directional position uncertainty model, the influence of unreliable position information is reduced.

5.3. Evaluation of sensor quality and particle numbers

In the next scenario as shown in Fig. 5(b), the position accuracy of SPBP, BPUCL and proposed algorithm under the impact of various inertial sensor bias (i.e., ϵ and ∇) is analyzed. The scenario condition is set as follows: UAVs are cruising toward north, UAV 2–5 are equipped with the same grade of INS as depicted in Table 1, while UAV 1 is equipped with different grade of INS (shown in the abscissa axis). The performance under the influence of various INS settings is shown in Fig. 8, and the impact of particle numbers on position accuracy is analyzed in Fig. 9. The following two observations can be made:

(1) The RMSE of the three algorithms all increase with INS error, while the proposed algorithm maintains the best navigation accuracy. Similar as the previous scenario, the cause for poor performance of SPBP also lies in not taking accumulated errors into consideration. As for BPUCL, although the accumulated error is considered, the weights of particles are calculated by the means of importance sampling and received information are treated equally, resulting in introducing negative impact of unreliable information in the fusion process and thus higher errors than the proposed algorithm. It should be mentioned that the factor about time interval of GNSS signal lost has the same property.

(2) It can be observed that the RMSE of both methods are dropped as the number of particles increase. This is due to the fact that the algorithm performance is highly sensitive to the quality of prediction, and the probability of true position value covered would rise with the number of particles. Moreover, the proposed method outperforms BPUCL given the same number of particles. The main reason is that, compared to BPUCL, the proposed WUBP method can achieve higher density because of utilizing the directional position uncertainty model.

Remark 1. These observations of different simulation scenarios indicate that the entire group can reap the benefits of accurate positional estimate from UAVs with less maneuvers, higher-grade of INS or shorter time interval of GNSS signal lost, which is a valid proof for navigation performance improvement. In addition, these results may provide the insights into the connection between formation control schemes and decentralized swarm localization algorithm.

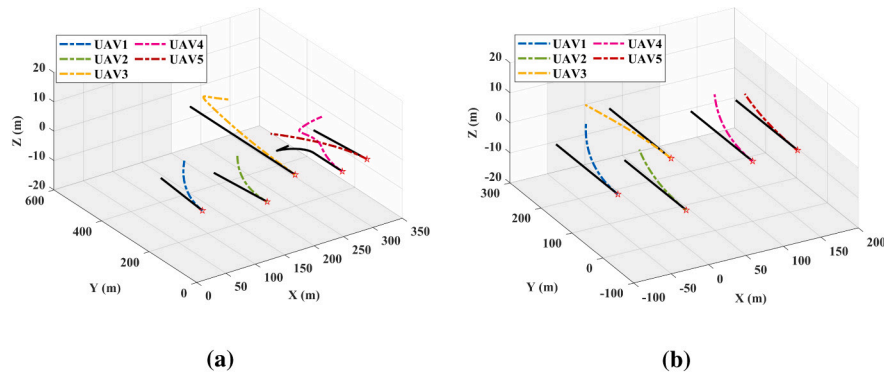


Fig. 5. The trajectories related to UAVs in one simulation run. Colored and black curves represent the INS and true trajectories, respectively.

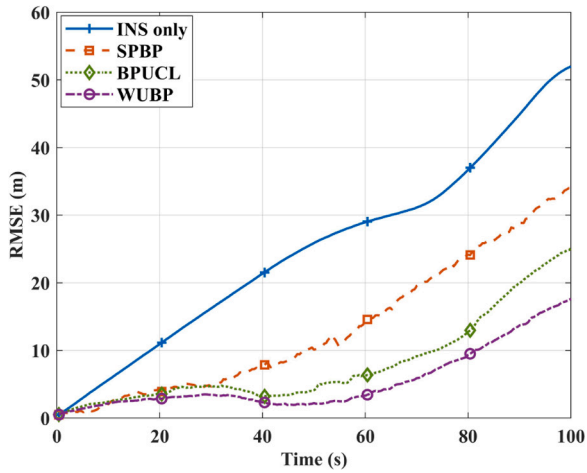


Fig. 6. Positioning error at different timestamps.

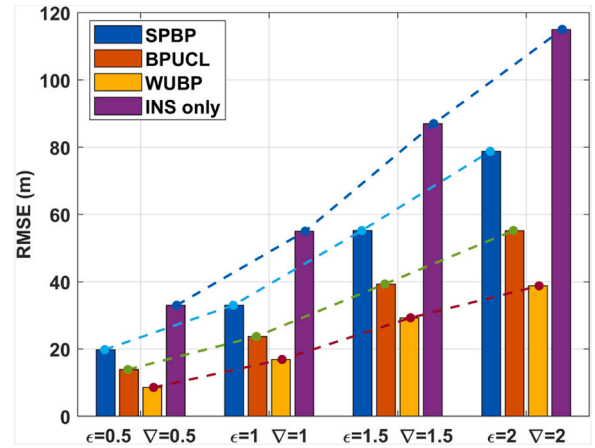


Fig. 8. Analysis of the performance under the influence of various INS settings.

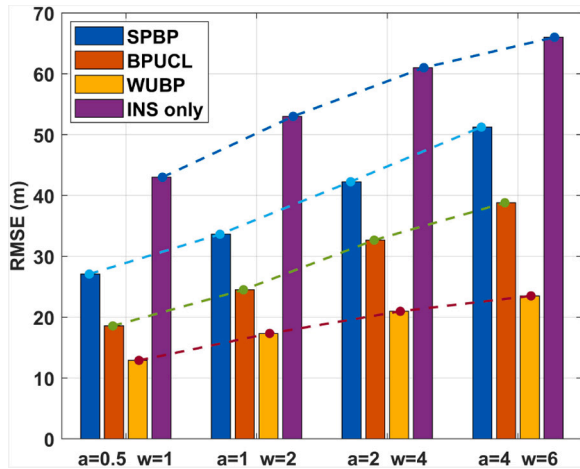


Fig. 7. Evaluation of the position accuracy under the influence of UAV dynamic maneuvers.

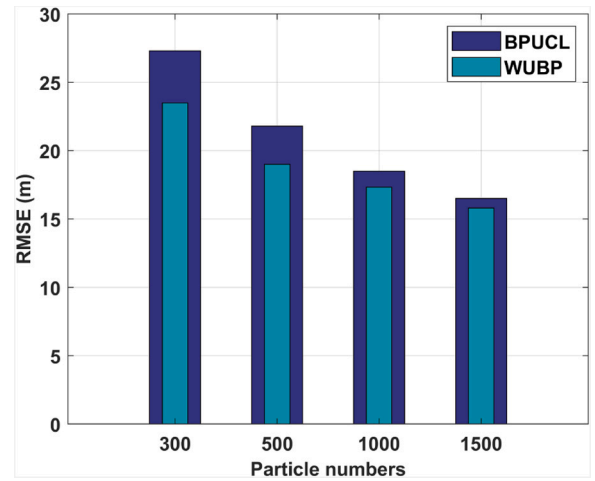


Fig. 9. Evaluation of the impact of particle numbers on the position accuracy.

6. Real world experiments

In this section, we present a real world experiment with simultaneous moving unmanned vehicles to reveal the superior performance of the proposed WUBP algorithm.

6.1. System configuration

As shown in Fig. 10, onboard IMU YIS370-U¹ is used to perform dead reckoning with strapdown inertial navigation algorithm. The sensor is integrated on vehicle board. In addition, we equipped each

¹ www.yesense.com/yis370.

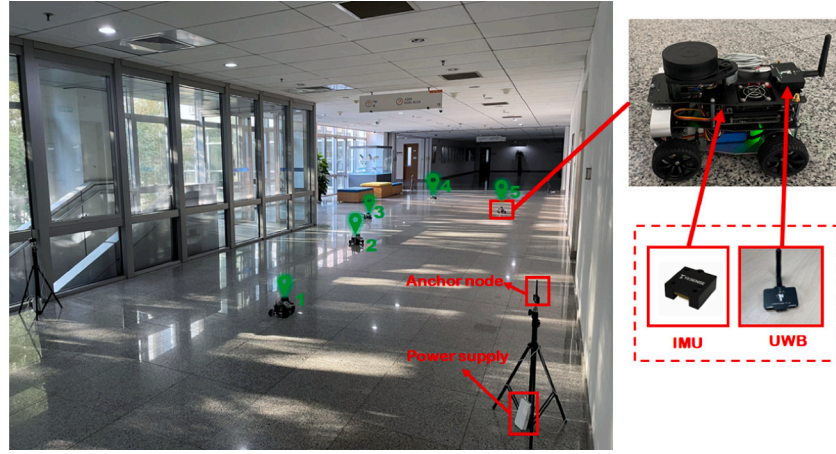


Fig. 10. Data collection in the building and sensors carried by a vehicle.

Table 2

Experimental results for position error.

Method	INS only	WUBP	BPUCL	SPBP
Position error	8.69 m	6.58 m	7.33 m	7.88 m
Improvement	–	24.3%	15.6%	9.3%

unmanned vehicle with UWB transceiver from Noooploop,² which has been securely mounted on the vehicle. The UWB system is capable of performing communication while providing peer-to-peer range measurements with an accuracy of 10 cm. The data were collected in real-time and stored on the embedded board of vehicles, which includes INS outputs for all vehicles (100 Hz), predicted beliefs from data transmission (10 Hz) and inter-vehicle ranges (10 Hz). These data sets were then post processed in MATLAB to evaluate the algorithms. Based on the sensor characteristics, the parameters are set as follows: $\epsilon = 5^\circ/\text{h}$, $\nabla = 0.5 \text{ mg}$, $\sigma_{(\delta w)} = 0.5^\circ/\text{h}$, $\sigma_{(\delta f)} = 0.1 \text{ mg}$, $M = 1000$.

The real world experiment has been performed in the university building. As depicted in Fig. 11, six UWB transceivers anchored at the hallway to provide ground truth in the indoor environment. The UWB transceivers are placed on the same plane with tripods, and the indoor local coordinate system is built by precisely measuring the distance between them. Once the communication between the UWB transceivers and the laptop was established, all the vehicles began to move from outside of building to inside. This design means to simulate the robots enter a GNSS-denied environment. Note that the initial information of position and orientation can be obtained from INS/GPS integrated navigation mode while vehicles are moving outside of the building. To verify the performance of cooperative navigation under dynamic conditions, five set of tests were conducted with different maneuvers. The path of different vehicles is shown in Fig. 11. Vehicle 1 and vehicle 2 are programmed with the robot operating system (ROS) to smoothly move in straight lines and enter the building relatively late, which are considered to be nodes with higher position information intensity. The other vehicles are programmed to move with several different maneuvers. Each vehicle concurrently performs ranging and communication.

6.2. Experimental results

The performance is evaluated by collecting the trajectory of each vehicle and plotting the 2- σ confidence ellipses as shown in Fig. 12, and the results for the absolute position errors is shown in Table 2.

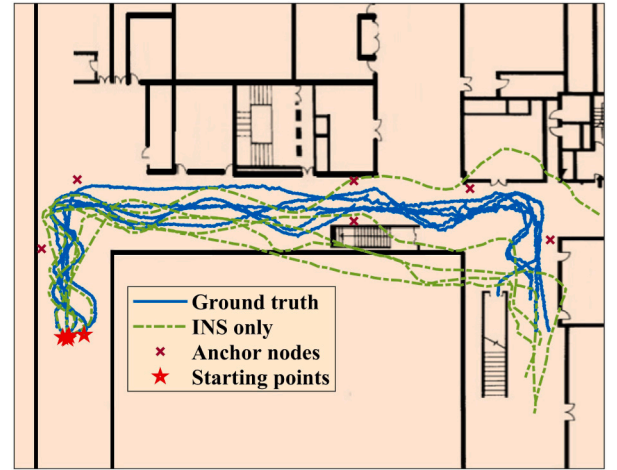


Fig. 11. The path of different vehicles and experimental setup.

As can be seen in Fig. 12(a), SPBP has the worst performance both in the x direction and y direction. Meanwhile, WUBP has the smallest axis of confidence ellipse among the three algorithms and achieves a positioning accuracy of 6.58 m. In addition, Fig. 12(b) visualizes the trajectory of vehicle with maneuvering. By incorporating the proposed position uncertainty model and position information intensity in the cooperative navigation process, WUBP can follow the maneuvering track of the vehicle and significantly improves the position accuracy in comparison to BPUCL method.

Fig. 13 illustrate the RMSE of position estimates with respect to different methods. Similar to the findings in the simulation, the advantages of WUBP can be clearly seen in the case when vehicles performing maneuvers, in which the vehicle 3, 4 and 5 have greater position errors.

6.3. Discussion

The results of real world experiments indicate that the proposed WUBP method gives the best localization accuracy and is more robust with vehicle maneuvers. The reason for this can be explained as follow: since the directional position uncertainty model can approximate the associated uncertainty of position estimates with different influence factors, the particles in the proposed WUBP algorithm may precisely cover the real position and alleviate the impact of dynamic maneuvers. In addition, owing to the utilization of the position information intensity, the weights of navigation information from different vehicles can be tuned

² www.nooploop.com.

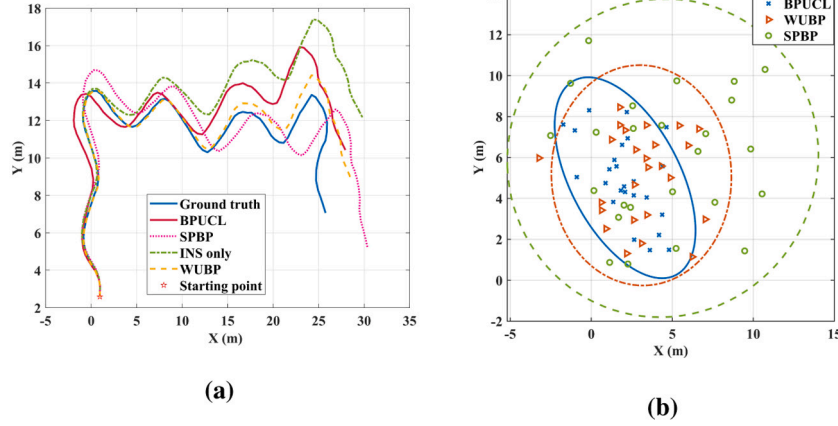


Fig. 12. The trajectory (vehicle 3) of different methods and the error confidence ellipses from indoor navigation experiment.

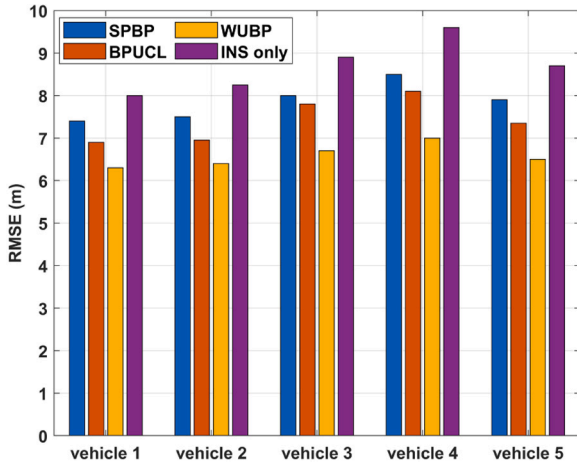


Fig. 13. The RMSE of position estimates against different vehicles.

adaptively based on single vehicle navigation performance, making the whole system receive more benefit from cooperative navigation.

Furthermore, since predicting upper bounds for the position errors in real world is more challenging, the performance of WUBP algorithm is poor when compared to the result in the simulation. In particular, the errors of IMU are highly instable due to the quality of MEMS inertial sensor, leading to a decrease in the accuracy of estimated uncertainty. Thus, a better position accuracy can be achieved if we use higher grade IMU sensors.

7. Conclusion

In this paper, we propose the WUBP algorithm to alleviate the impact of accumulated error induced by INS and enhance the performance of the cooperative navigation system in GNSS-denied areas. First, we derived the directional position uncertainty model under dynamic conditions to characterize the position estimations, providing insights into the relationship between the uncertainty of position estimation and effective factors. Second, we designed the position information intensity to reflect the quality of position estimate and tune the weights of incoming messages for information fusion. The experimental results show that WUBP outperforms the state-of-the-art BP-based methods in terms of accuracy, and 24.3% precision improvement can be achieved. Besides, the performance of our method is fundamentally limited due to the instable random noise from inertial sensor and the short-time approximation in the analytical expressions of position error, resulting in an accuracy degrade with a long flight.

One of future work is to combine with a path planning strategy and to design a cooperative guidance algorithm that commands specific maneuvers to further limit position estimation error growth. Another interesting future work would be the incorporation with map-aided, visual and magnetic field-based methods for a more robust and comprehensive GNSS-denied cooperative navigation strategy.

CRediT authorship contribution statement

Jian Li: Conceptualization, Methodology, Software, Writing – original draft. **Gongliu Yang:** Supervision. **Qingzhong Cai:** Visualization, Investigation, Reviewing and editing. **Haofei Niu:** Data curation. **Jing Li:** Software, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

We consider the navigation frame (n-frame) as the East-North-Up (E-N-U) geography frame. Let i , b and e denote the inertial frame, the body frame and the earth frame respectively. The nonlinear error differential equations of INS can be formulated as:

$$\dot{\phi} = \phi \times (\omega_{ie}^n + \omega_{en}^n) + \delta\omega_{in}^n - C_b^n \delta\omega_{ib}^b \quad (A.1a)$$

$$\delta\dot{\mathbf{v}}^n = \mathbf{f}_{sf}^n \times \phi + \mathbf{v}^n \times (2\delta\omega_{ie}^n + \delta\omega_{en}^n) - (2\omega_{ie}^n + \omega_{en}^n) \times \delta\mathbf{v}^n + C_b^n \delta\mathbf{f}_{st}^b \quad (A.1b)$$

$$\delta\dot{\mathbf{p}} = \begin{bmatrix} 0 & 1/R & 0 \\ \sec L/R & 0 & 0 \end{bmatrix} \delta\mathbf{v}^n \quad (A.1c)$$

where $\delta\mathbf{v}^n$, ϕ and $\delta\mathbf{p}$ are respectively the error vector of velocity, attitude and position in n-frame, C_b^n denote the transformation matrix from the b-frame to the n-frame, ω_{ie}^n is the rotational angular velocity of the earth, ω_{en}^n is the angular velocity of the vehicle relative to the earth, $\delta\omega_{ie}^n$ and $\delta\omega_{en}^n$ represent the corresponding errors, R is the mean radius of the Earth, $\mathbf{f}_{sf}^n = [f_E, f_N, f_U]^T$ is acceleration in n-frame, L

is the latitude, δf_{st}^b and $\delta \omega_{ib}^b$ are measurement error of accelerometer and gyroscope.

For inertia sensors used in UAV, the magnitude of angular velocity and acceleration error induced by carrier maneuver is much larger than that of Earth rotation rate as well as the second and third terms on the righthand side of Eq. (A.1b). Thus, these terms can be safely ignored. In view of the above considerations, the complete and complex system error differential Eq. (A.1) can be simplified and summarized in a compact form as:

$$\dot{X} = FX + B \quad (A.2)$$

where $X = [\phi_E \ \phi_N \ \phi_U \ \delta v_E \ \delta v_N \ \delta p_N \ \delta p_E]^T$ denote the UAV error state vector, and $B = [-\delta \omega_E \ -\delta \omega_N \ -\delta \omega_U \ \delta f_E \ \delta f_N \ 0 \ 0]^T$ is the sensor error in n-frame. The coefficient matrix F can be rewritten as:

$$F = \begin{bmatrix} 0_{3 \times 3} & M_1 & 0_{3 \times 2} \\ M_2 & 0_{2 \times 2} & 0_{2 \times 2} \\ 0_{2 \times 3} & M_3 & 0_{2 \times 2} \end{bmatrix}$$

$$\text{where } M_1 = \begin{bmatrix} 0 & -1/R \\ 1/R & 0 \\ \tan L/R & 0 \end{bmatrix}, M_2 = \begin{bmatrix} 0 & -g & f_N \\ g & 0 & -f_E \end{bmatrix}, M_3 = \begin{bmatrix} 0 & 1/R \\ 1/R & 0 \end{bmatrix}.$$

For tractability of analysis, we neglect the Foucault frequency and the coupling term of earth rotation velocity, then the time-domain solution of simplified system position error differential equation can be expressed as a Gauss–Markov process:

$$\delta p(t) = \phi(t)X(0) + \int_0^t \phi(t-\tau)B d\tau \quad (A.3)$$

$$\text{where } \begin{bmatrix} R(1 - \cos \omega_s t) & 0 & \frac{f_E R}{g}(\cos \omega_s t - 1) & 0 & \frac{\sin \omega_s t}{\omega_s} & 1 & 0 \\ 0 & R(1 - \cos \omega_s t) & \frac{f_N R}{g}(1 - \cos \omega_s t) & \frac{\sin \omega_s t}{\omega_s} & 0 & 0 & 1 \end{bmatrix},$$

$\omega_s = \sqrt{g/R}$ is the Schuler angular frequency, $X(0)$ is the initial error state.

Generally, the sensor error often incorporates deterministic as well as stochastic components. For the purpose of characterizing the position error accumulation, consider the sensor error as constant or stationary random process respectively, then the mean and covariance of position errors $\delta p(t)$ can be expressed as:

$$E\{\delta p(t)\} = \phi(t)E\{X(0)\} + \int_0^t \phi(t-\tau)B d\tau \quad (A.4a)$$

$$D\{\delta p(t)\} = \phi(t)P\{X(0)\}\phi^T(t) + \int_0^t \phi(t-\tau)B(\tau)B^T(\tau)\phi^T(t-\tau)d\tau \quad (A.4b)$$

where $P\{X(0)\}$ and $E\{X(0)\}$ denote the mean vector and the covariance matrix of initial error state.

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